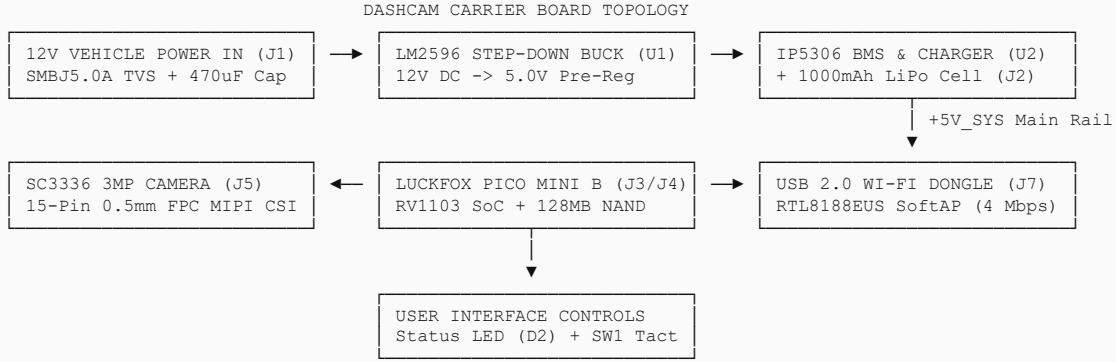
CARRIER BOARD SPECIFICATION: HARDWARE ARCHITECTURE & NET SCHEDULE

1. CARRIER BOARD DESIGN OBJECTIVE & FUNCTIONAL SUBSYSTEMS

The Dashcam Carrier Board (ENG-DASH-PCB-001) serves as the central electro-mechanical backplane integrating the **Luckfox Pico Mini B SBC (RV1103)**, **SmartSens SC3336 3MP Camera**, **1000mAh LiPo Battery**, **IP5306 BMS Module**, and automotive power conditioning.



2. SUBSYSTEM NET INTERCONNECT & DECOUPLING MATRIX

SUBSYSTEM BLOCK	REF DES	COMPONENT / CONNECTOR	KEY SIGNALS / POWER RAILS	DECOUPLING & PROTECTION
Automotive In	J1, D1, C1	Terminal Block 2-Pin 5.08mm	+12V_IN, GND	SMBJ5.0A TVS + 470uF/25V Electrolytic Reservoir
Pre-Regulator	U1, C2, C3	LM2596 DC-DC Step-Down Module	+12V_IN -> +5V_PRE (Pre-regulated 5V)	470uF/25V Reservoir + 100nF Ceramic Decoupling
BMS & Battery	U2, J2, R1, C4, C5	IP5306 Synchronous PMIC + JST-PH	VBAT (+3.7V LiPo), +5V_PRE In, +5V_SYS Out	R1 (100 Ohm) Keep-Alive Dummy Load + 100nF Decoupling
SBC Sockets	J3, J4, C6, C7	2x 1x08 2.54mm Female Sockets	+5V_SYS, +3V3, UART2, I2C3, USB, GPIOs	100nF MLCC at +5V_SYS (J4) and +3V3 (J3)
Camera MIPI	J5, C8	15-Pin 0.5mm FPC Bottom-Contact	MIPI_D0_P/N, MIPI_CLK_P/N, MIPI_D1_P/N, SCCB	100nF Decoupling on Sensor 3.3V Supply Rail
Wi-Fi Dongle	J7, C9	USB Type-A Female Receptacle	+5V_SYS (VBUS), USB_DM (D-), USB_DP (D+), GND	100nF MLCC Decoupling directly across VBUS/GND
User Interface	SW1, R2, D2, R3	Tactile Switch + 0805 Green LED	SW_SENSE (GPIO34), STAT_LED (GPIO48)	R2 (10k Pull-up), R3 (470 Ohm LED Current Limiting)

ORGANIZATION / PROJECT: DASHCAM PROTOTYPE PROJECT	SIZE: A4	DOCUMENT NUMBER: ENG-DASH-PCB-001	DATE: 2026-10-08	SHEET: 1 OF 2	
DOCUMENT TITLE: CARRIER BOARD HARDWARE SPEC	REV: 1.4	DESIGN ENGINEER: Souvik H	ENGINEER REGISTRATION NO.: IN-ENG-2022-22824		
		ENGINEERING DISCIPLINE: EMBEDDED HARDWARE & PCB DESIGN	STATUS: RELEASED		

CARRIER BOARD SPECIFICATION: PINOUT SCHEDULES & PCB LAYOUT RULES

3. COMPREHENSIVE CONNECTOR PINOUT SCHEDULE

CONNECTOR	PIN NUMBER	NET DESIGNATION	SIGNAL STANDARD	FUNCTIONAL DESCRIPTION
J1 (12V Input)	Pin 1 / 2	+12V_IN / GND	Automotive 12V DC	Vehicle ignition / accessory electrical line entry.
J2 (LiPo Batt)	Pin 1 / 2	VBAT / GND	Lithium Polymer (3.7V - 4.2V)	1000mAh battery cell buffer with integrated PCM.
J3 (SBC Left)	Pins 1 to 8	UART2_TX, RX, GND, +3V3, I2C3_SCL, SDA, GND, STAT_LED	LVCMOS 3.3V Logic	Console debug, peripheral power, camera SCCB, LED drive.
J4 (SBC Right)	Pins 1 to 8	+5V_SYS, GND, USB_DM, USB_DP, CAM_RSTn, CAM_PDN, SW_SENSE, GND	USB 2.0 / DC Power	Main 5V system entry, USB host signals, camera control.
J5 (Camera FPC)	Pins 1 to 15	GND, D0_N/P, GND, CLK_N/P, GND, D1_N/P, GND, RST, PDN, SCL, SDA, 3V3	MIPI D-PHY (800 Mbps/lane)	SmartSens SC3336 3MP raw Bayer image sensor link.
J7 (Wi-Fi USB)	Pins 1 to 4	+5V_SYS, USB_DM, USB_DP, GND	USB 2.0 High-Speed	RTL8188EUS 802.11n Wi-Fi adapter receptacle.

4. HIGH-SPEED PCB LAYOUT GUIDELINES & PHYSICAL CONSTRAINTS

• **MIPI CSI-2 Differential Pairs:** Route MIPI_D0, MIPI_CLK, and MIPI_D1 as 100 Ohm differential controlled impedance microstrips over an unbroken solid ground plane (Layer 2). Keep intra-pair length skew below 0.5 mm (20 mils) and inter-pair skew below 1.5 mm.

• **USB 2.0 Differential Routing:** Route USB_DM and USB_DP as 90 Ohm differential pairs. Avoid via transitions between SBC and USB-A receptacle.

• **Power Distribution:** Form a wide polygon pour for +5V_SYS and +12V_IN rails (>35 mil trace width or plane pour) to minimize IR drop during Wi-Fi transmission bursts (up to 805 mA peak system draw).

• **Physical Dimensions & Mounting:** Form factor: 60.0 mm x 38.0 mm, 2-layer FR4, 1.6 mm thickness, 1 oz copper. Features 4x M2.5 mounting holes on a 53.0 mm x 31.0 mm rectangle matching the enclosure standoffs.

• **Manufacturing Gerbers & 3D Renders:** Production Gerbers (F_Cu, B_Cu, F_SilkS, B_SilkS, F_Mask, B_Mask, Edge_Cuts, Drill) exported to `hardware/source/carrier_board/gerbers/`. High-resolution raytraced 3D renders available in `hardware/source/carrier_board/carrier_board_3d_top.png` and `carrier_board_3d_iso.png`.

• **KiCad 10 Project Assets:** Fully editable KiCad project files located in `hardware/source/carrier_board/` (`carrier_board.kicad_pro`, `carrier_board.kicad_sch`, `carrier_board.kicad_pcb`, `carrier_board_schematic.pdf`, `carrier_board_top_layer.pdf`, `carrier_board_bottom_layer.pdf`).

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			DESIGN ENGINEER Souvik H	ENGINEER REGISTRATION NO. IN-ENG-2022-22824	
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